

Passivization by Voice: A Merge-Based Approach

Chigchi Bai

Inner Mongolia University of Finance and Economics

Keywords: passivization, voice, subject, argument, merge

1. Introduction

At least two syntactic views on how passive semantics is assigned are available in the generative literature. On the mainstream (Chomskyan) view, passive semantics is configurationally assigned by promoting the patient and suppressing the agent. By contrast, it was recently proposed that passive semantics is assigned by a dedicated Voice head that does not project agents (Embick 2004b, Schäfer 2008b, Bruening 2012, Harley 2013, Legate 2014, Alexiadou et al. 2015, Legate et al. 2020). I label such a head and its equivalents “–D Voice”. –D Voice was postulated in parallel with Kratzer’s (1996) Voice, which introduces the external argument (agent). –D Voice has nothing to do with passivization as a syntactic operation, yet it was assumed to contribute to producing a passive meaning. On the mainstream (Chomskyan) view, however, passivization as a syntactic operation goes hand in hand with producing a passive meaning.

This controversy, as we will see in this paper, is due to the failure to correctly identify passivization as successive-cyclic movement and correctly characterize Voice as a syntactic head introducing (potential) subjects. Ever since Chomsky (1981), it has been assumed that passivization is one-step movement of the patient to the surface subject position. In this paper, I demonstrate that the subject of passives stops over in an intermediate position before reaching the NOM position.

2. Evidence for passivization as a short movement

I use the informal term “short movement” to refer to the object’s movement that targets Spec of a head in the voice domain and “long movement” to refer to its movement that targets Spec of T. In this section, I present evidence that passivization is a short movement, rather than a long movement.

(1) Passivization as a short movement: $[_{TP} [_{VoiceP} DP [_{VP} t_{DP}]]]$

(2) Passivization as a long movement: $[_{TP} DP [_{VoiceP} [_{VP} t_{DP}]]]$

Evidence comes from Japanese causative-of-passive sentences (passive embedded under causative) in which the patient of the verb stem occurs as a causee and is assigned dative case (DAT). Obviously, causative-of-passive sentences such as (3) are derived out of a passive phrase, as discussed by Saito (1982: 92) and Aoyagi (2021: 99).

- (3) Ziroo-ga Hanako-o/ni Taroo-ni sikar-are-sase-ta.
 Ziro-NOM Hanako-ACC/DAT Taro-DAT scold-PS-CS-PST
 ‘Ziro made Hanako be scolded by Taro.’ (Tsujimura 1996: 259)

In (3), the passive phrase is not directly embedded under T, which is spelled out by *-ta*, but by a causative head, which is spelled out by *-sase*. A passive phrase can certainly be embedded under T, as in (4).

- (4) Hanako-ga Taroo-ni sikar-are-ta.
 Hanako-NOM Taro-DAT scold-PS-PST
 ‘Hanako was scolded by Taro.’ (Tsujimura 1996: 258)

Note that the patient *Hanako* in (4) is not base-generated in its surface position since that position is not thematic, but within the thematic domain vP as an object.

- (5) ... Hanako_i ... [_{VP} Taroo [_{VP} *t_i* sikar] -are] ...

The same holds true in (3), in which the surface position of *Hanako* cannot be thematic. If it were a theta position, *sikar-are* ‘be scolded’ would be a complex intransitive verb, which assigns a theta role to *Hanako* as a thematic subject, and *-are* would not be a passive morpheme but part of the verb stem. However, as a fact, *sikar-are* is not an intransitive verb but the passive form of the transitive verb *sikar* ‘scold’, as evidenced by its ability to occur with the agentive *by*-phrase *ni yotte*.

- (6) Ziroo-ga Hanako-o/ni Taroo-ni yotte sikar-are-sase-ta.
 Ziro-NOM Hanako-ACC/DAT Taro-to owing scold-PS-CS-PST
 ‘Ziro made Hanako be scolded by Taro.’ (Hoshi 1999: 208)

The surface position of the non-NOM subject *Hanako* is not Spec of a non-finite TP. If it were, *Hanako* would not be differentially marked by DAT (*-ni*) and ACC (*-o*), given that only alternation of NOM (*-ga*) and GEN (*-no*) is possible in Spec of TP in Japanese. In addition, in Japanese, which is an agglutinative language, causatives, syntactic or lexical, do not involve projection of embedded TP (Harley 2008, 2013); a causee always occupies Spec of a non-T head.

To conclude the above discussion, in (3), the non-NOM subject *Hanako* is base-generated

as an object and is promoted to Spec of a head lower than T, which is a NOM-assigner. This leads to the possibility that the surface subject (SBJ) of the sentence is in fact promoted from the same position as that where the embedded *Hanako* in (3) occurs, assuming that uniformity of derivation obtains between the passive phrases in (3) and (4). That is, passivizations of *Hanako* in (3) and (4) are identical operations. The structure of the uniform passive phrase is represented in (7).

(7) [_{XP} Hanako_i [_{VP} Taroo [_{VP} *t_i* sikar]] -are]

When XP, which is not TP, in (7) is selected by a causative head *-sase*, *Hanako* remains in Spec of XP, thereby yielding a causative-of-passive structure as in (3), and when XP is selected by T, *Hanako* further moves into Spec of TP, yielding an unembedded passive structure as in (4). Importantly, passivization is already completed by promoting *Hanako*, regardless of where it ends up in the surface structure. That is, passivization is a short step movement, not a long step one. This is exactly what Baker's (1985) Mirroring Principle (MP) predicts. Affixation of the passive morpheme *-are* and promotion of the patient by the head spelled out by *-are* are linked and mirror each other. The linking and mirroring fail if promotion of the patient targets Spec of TP directly, skipping over Spec of XP, because that promotion is not owing to X, which is spelled out by *-are*. Therefore, treating passivization as a one-step movement (Chomsky 1981 and much subsequent literature) is wrong, as correctly predicted by MP.

In what follows, let us elaborate on the morphology-syntax mirroring of A-to-D raising in causative-of-passive sentences. Notate the passive morpheme *-are* as PS and the causative one *-sase* as CS. Passivization takes place before causativization, which are both syntactic operations, as mirrored by the morphological string PS-CS. This means that promotion of *Hanako* must precede introduction of *Ziwoo*.

(8) [Ziwoo-ga [Hanako-o/ni [Taroo-ni [~~Hanako~~ sikar] -are] -sase] ...

Assume that to promote an element is to reintroduce (internally merge) it in a higher position. As predicted by MP, (re)introduction of the patient *Hanako* and the causer *Ziwoo* take place hand in hand with affixation of PS *-are* and CS *-sase*, respectively, as shown by the dotted lines. (Re)introduction of *Ziwoo* in Spec of TP (left out in (8)), which follows that of *Hanako* in Spec of XP, then takes place hand in hand with affixation of the tense morpheme *-ta*, which follows PS-CS. Obviously, one-to-one correspondences hold between (re)introduction of the arguments as syntactic operations and affixation of the morphemes in order.

Similarly, in (9), affixation of *-are* mirrors and bound with (re)introduction of *Hanako* in Spec of XP, whose head is *-are*. What mirrors (re)introduction of *Hanako* in Spec of XP is not affixation of the tense morpheme *-ta*, since *-ta* is not the exponent of X, but of T. Affixation of *-ta*, however, mirrors and is bound with (re)introduction (further promotion) of *Hanako* in Spec of TP.

- (9) [Hanako-ga [Taroo-ni [~~Hanako~~ sikar] -are] ...

On the one-step movement/long movement view of passivization, (re)introduction (further promotion) of *Hanako* would be mirrored and bound by the affixation of *-ta*, not by that of *-are*, rendering *-are* irrelevant with passivization of *Hanako*, contra the fact.

All this holds true of the case in which the causee, the passivized patient *Hanako* or *Dorž* in (10), is assigned ACC.¹ In (10), a Mongolian causative-of-passive sentence, the patient *Dorž* undergoes promotion and precedes the agent *aav*. As represented in (11), affixation of PS *-gd* is bound with reintroduction of *Dorž*, which represents passivization, and affixation of CS *-uul* with introduction of the causer (dropped in this sentence), which represents causativization.

- (10) Dorž-ig aav-d-ni tani-gd-uul-h-gui-in tuld sahal naa-san.
 Dorž-ACC father-DAT-RX recognize-PS-CS-INF-NEG-GEN for beard attach-PST
 ‘In order not to make Dorž recognized by his father, I attached beard to his face.’
 (Umetani 2006: 95)

- (11) [Causer [Dorž-ig [aav-d-ni [~~Dorž~~ tani] -gd] -uul] ...

To sum up this section, passivization is created by promoting the patient to a position higher than the agent, which is suppressed (case-downgraded), and, importantly, passivization that involves a NOM subject is successive-cyclic.

3. Introducing passive subjects through Voice

A consequence of the elaboration on the successive-cyclicality of passivization is that the subject of passives is introduced by an argument-introducer much as in active sentences. Given that promotion of the patient and suppression of the agent make up the core property of the passive, which is the voice proper, it is not deniable that a passivizing head can introduce an argument. This means that the passivizing head *X* in (7) is an argument-introducer like Kratzer’s (1996) Voice (or Chomsky’s 1995 *v*). Note that *X*, a passivizing head, internally merges (reintroduces) arguments and Voice, a transitivity (or causativizing) one, externally merges (first-introduces) them. Importantly, then, this yields no difference between the passivizing head *X* and the transitivity head Voice (roughly the same as *v* in (7)) with respect to their ability to introduce arguments. It then follows that the passivizing head *X* is also Voice, following Kratzer’s argumentation that Voice is an argument-introducer. Accordingly, the syntactic difference between actives and passives lies in the difference between external merge (EM) and internal merge (IM) and the height of the positions where EM and IM take place.

However, the passivizing Voice and the transitivity Voice are not the very same head; they are distinct instances of Voice instead. That is, VoiceP splits into two separate projections and the head of each (re)introduces the agent and the patient, respectively. Thus, the voice domain

is in fact a Voice-over-Voice configuration as in (12).

(12) ... [VoiceP2 Patient_i [VoiceP1 Agent [VP *t_i* ...

To rewrite (7) using (12), we get (13).

(13) [VoiceP2 Hanako_i [VoiceP1 Taroo [VP *t_i* sikar]] -are]

This leads to Nie's (2020) argumentation that causatives have a Voice-over-Voice structure. Combining Nie (2020) and (7), we arrive at the conclusion that causative-of-passive sentences have a three-layered Voice structure, in which the causer, e.g. *Ziroo* in (3) and the dropped subject in (10), is introduced by Voice3 via EM, as represented below.

(14) [VoiceP3 Causer [VoiceP2 Patient_i [VoiceP1 Agent [VP *t_i* ...

In (14), the passivizing head Voice2 is sandwiched between Voice1 and Voice3, which are both transitivizing heads. Importantly, all the separate heads split out of Voice are argument-introducers. In this sense, VoiceP is a split projection much like CP, which splits into separate projections such as FocP and TopP.

If the Split VoiceP analysis is on the right track, it predicts that it is always the highest argument (the last-merged one) that is promoted to the NOM subject (SBJ), regardless of whether the sentence is active/causative or passive. This is not predicted by the mainstream one-step movement approach to passivization. It then naturally follows that SBJ of both active and passive sentences is a Voice-internal subject. Much as vP-internal (VoiceP1-internal) subjects of actives, subjects of passives are internal to VoiceP2, where they are assigned subjecthood. That is, all arguments are potential subjects (Sbj). In the course of derivation, clauses can have more than one sbj introduced by Voice but only one SBJ. The the last-merged sbj (the highest one) becomes SBJ.

This leads us to the conclusion that Voice is not merely an agent-introducing head; it is instead a sbj-introducing head, sbj being either an agent or a patient. The split property of Voice connects to its ability to derive voice alternants including passives and causatives. That is, Voice is the single engine that syntactically derives voice alternations. This challenges previous approaches that were proposed in favor of division between argument-introducing Voice (+D Voice) and -D Voice.

Since the passive, as one of the voice alternants, does not necessarily realize an overt agent, quite many studies including Embick (2004b), Schäfer (2008b), Bruening (2012), Harley (2013) and Alexiadou et. al. (2015) have assumed agent-less/-D Voice in various flavors. They, however, are explicitly criticized by Collins (2024), who presents empirical evidence for and theoretical reasoning to the position that external arguments, whether overt or implicit, are syntactically projected in passives.² Collins' (2024) theoretical reasoning is based on his

Argument Criterion, which is presented in parallel with Chomsky's (1981: 36; 1986: 97) Theta-Criterion.

(15) Argument Criterion (AC)

- a. Each argument is introduced by a single argument-introducing head.
- b. Each argument-introducing head introduces a single argument. (Collins 2024: 8)

One effect of Argument Criterion is that it prevents an argument-introducing head from introducing no argument because it forces a bijection (one-to-one relation) between argument-introducing heads and arguments (Collins 2024: 9). Incorporating Argument Criterion into the Split VoiceP analysis brings about the following condition.

(16) Anti-Vacuous Condition (AVC)

A morphologically overt Voice head does not project a semantically vacuous Spec.

AVC, being well compatible with AC, predicts that no head with voice semantics (i.g. passive, causative, etc.) exists without introducing sbj, contra the –D Voice view. According to the –D Voice view, passive and causative semantics is predetermined by dedicated voice-specifying features like [PASSIVE] and [ACTIVE]/[CAUSE] postulated for a Voice-like head. However, a feature-based creation of passive and causative semantics is in fact a last-resort and not preferred since such features, unlike [CASE], [ϕ] and [FINITE/TENSE], which are all primitive, are not well-motivated. Chomsky's (1981) classic analysis predicts that a passive meaning is produced by a configuration in which the patient is promoted skipping over the agent, but not produced by postulating semantic features on formal heads. Unfortunately, the classic one-step movement analysis, though it is rooted too deep to cast off from sentence syntax and word syntax, faces empirical problems, not regarding passive semantics though.

The crux of the matter in dealing with the dilemma in the classic analysis and the controversy between it and the lately developed –D Voice analysis is to abandon the deep-rooted one-step tradition and to find an alternative way to account for what was intended by Chomsky (1981), without violating the conceptual standards such as the severing of the external argument from the verb and making nominative subjects originate as potential subjects invariably in actives and passives. As to the technical machinery to use for this purpose, –D Voice should be abandoned, too, for the above mentioned reasons.

Taking external and internal arguments to be instances of sbj as a conceptual standard, the Split VoiceP analysis accounts for the problems with both the one-step movement approach and the –D Voice approach. On the Split VoiceP view, passive semantics is assigned by the Voice-over-Voice configuration.³ That is, introducing an argument through Voice via IM produces passive semantics.⁴

4. Conclusion

In this paper, I have argued for the claim that arguments are introduced as potential subjects by Voice, which is the single syntactic head endowed with the argument-introducing ability in the universal inventory of functional elements, echoing Kratzer's (1996: 120) statement that Voice is truly at the heart of a theory of voice. To summarize the discussion in this paper:

- a. Passivization is created by internally merging the patient argument through Voice;
- b. The NOM subject of passives as well as that of actives is a Voice-internal subject in that it is introduced beforehand by Voice to get assigned subjecthood;
- c. The voice domain is a Split VoiceP structure and clauses are built by introducing (a) potential subject(s) through separate Voice heads and promoting a last-merged one to the NOM position, with others being suppressed or demoted;
- d. Voice is not merely an external-argument-introducing head; it is a potential-subject-introducing head;
- e. Voice as a syntactic head is the single engine that manipulates voice alternants such as passives and causatives;
- f. Passive and causative semantics is assigned by the Voice-over-Voice configuration of Split VoiceP;
- g. No dedicated heads such as Passive, Voice_[passive], Voice_[active], Cause, v and their equivalents are necessary, nor are dedicated voice-specifying features;
- h. Introduction of arguments comes down to the simplest operation Free Merge (Chomsky 2013, 2015): (Re)merger of an argument, external or internal, is unconstrained; UG requires just this much for voice phenomena.

NOTES

¹ The differential case-marking of the causee in Japanese causatives, including causative-of-passive sentences, seems to have to do with semantic factors such as affectedness, volition, animacy, and so on; it is not purely syntactic.

² Surprisingly, however, Collins (2024) refutes Kratzer (1996) altogether with those who advocate Inert Voice, by introducing agents through v, a head lower than Voice and stating that "VoiceP has nothing to do with projecting the external argument, but is rather implicated in how the argument DPs are realized in A-positions" (Collins 2024: 109). However, Collins' (2024) separation of Voice from its argument-introducing ability remains problematic because if agents were introduced by v, not by Voice, they would not be argument severed from the verb. Note that Voice, in Kratzer's (1996) theory, is the lowest head that can sever the external argument from the verb.

³ This paper adopts the following definition of passive subject.

- (i) If DP instantiates DP(λy (P y, x, t_y)), where y is lower than x, x being non-nominative, in the thematic hierarchy with respect to the predicate P, then P is passive and DP is subject of P.

"Passive semantics" then refers to the meaning interpreted for such a configuration. On a –D Voice view, it would be the case that what solely contributes to producing the passive meaning

is the agent-suppressing property of the –D voice head, or a dedicated voice-specifying feature like [passive/non-act] on it. That is, syntax is not truly autonomous in producing the passive meaning on a –D Voice view, which would not specify a configuration such as “(P x, y)”, appealing only to non-nominative y, for deriving passives.

⁴ Introducing an argument via EM produces causative semantics. Also see Nie (2020: 105, 115), who demonstrates that causative semantics is assigned configurationally.

REFERENCES

- Aoyagi, Hiroshi (2021) “On the Causative and Passive Morphology in Japanese and Korean,” *Open Linguistics* 7: 87-110.
- Alexiadou, Artemis, Elena Anagnostopoulou and Florian Schäfer (2015) *External Arguments in Transitivity Alternations: A Layering Approach*, Oxford University Press, Oxford.
- Baker, Marc (1985) “The Mirror Principle and Morphosyntactic Explanation,” *Linguistic Inquiry* 16: 373-415.
- Bruening, Benjamin (2012) “By-phrases in Passives and Nominal,” *Syntax* 16: 1-41.
- Chomsky, Noam (1981) *Lectures on Government and Binding*, Foris, Dordrecht, The Netherlands.
- Chomsky, Noam (1986) *Knowledge of Language: Its Nature, Origin, and Use*, Praeger, New York, NY.
- Chomsky, Noam (1995) *The Minimalist Program*, MIT Press, Cambridge, MA.
- Chomsky, Noam (2013) “Problems of Projection,” *Lingua* 130: 33-49.
- Chomsky, Noam (2015) “Problems of Projection: Extensions,” *Structures, Strategies and Beyond: Studies in Honour of Adriana Belletti*, ed. by Elisa Di Domenico, Cornelia Hamann and Simona Matteini, 3-16, John Benjamins, Amsterdam.
- Collins, Chris (2024) *Principles of Argument Structure: A Merge-Based Approach*, MIT Press, Cambridge, MA.
- Embick, David (2004b) “On the Structure of Resultative Participles in English,” *Linguistic Inquiry* 35: 355-92.
- Harley, Heidi (2008) “On the Causative Construction,” *Handbook of Japanese Linguistics*, ed. by Miyagawa, S. and Saito, M., 20-53, Oxford University Press, Oxford.
- Harley, Heidi (2013) “External Arguments and the Mirror Principle: On the Distinctness of Voice and v,” *Lingua* 125: 34-57.
- Hoshi, Hiroto (1999) “Passives,” *The Handbook of Japanese Linguistics*, ed. by Natsuko Tsujimura, 191-235, Blackwell, Malden MA.
- Kratzer, Angelika (1996) “Severing the External Argument from Its Verb,” *Phrase Structure and the Lexicon*, ed. by J. Rooryck and L. Zaring, 109-137, Kluwer, Dordrecht.
- Legate, Julie A (2012) “The Size of Phases,” *Phases: Developing the Framework*, ed. by Ángel J. Gallego, 233-250, Mouton de Gruyter, Boston, MA.
- Legate, Julie A (2014) “Voice and v: Lessons from Acehnese,” MIT Press, Cambridge, MA.
- Legate Julie A., Faruk Akkuş, Milena Šereikaitė, Don Ringe (2020) “On Passives of Passives,”

Language 96: 771-818.

Nie, Yining (2020) “Morphological Causatives are Voice over Voice,” *Word Structure* 13: 102-126.

Saito, Mamoru (1982) *Case Marking in Japanese*. Unpublished manuscript, MIT Press, Cambridge, MA.

Schäfer, Florian (2008b) *The Syntax of (Anti-)Causatives: External Arguments in Change-of-state Contexts*, John Benjamins, Amsterdam.

Shibatani, Masayoshi (1976) “The Grammar of Causative Constructions: A Conspectus,” *Syntax and Semantics: The Grammar of Causative Constructions*, ed. by Masayoshi Shibatani, 1-39, Academic Press, London.

Tsujimura, Natsuko (1996) *An Introduction to Japanese Linguistics*, Blackwell, Malden, MA.

Umetani, Hiroyuki (2006) “Serial Use of the Passive and Causative Suffixes in Mongolian,” *Gengo kenkyuu no shyatei*, ed. by Kato Shigehiro and Yoshida Hiromi, 83-102, Sanbi Insatsu, Tokyo.